

BIM216 – Database Management Systems

Term Project

For the term project, you are required to work as a group of two. Please note that you must continue with the group project (topic) that is assigned to you at the beginning of the semester, thus you are not allowed to change your topic.

Due Dates of Project Deliverables

Due Date for “Progress Report” – **April, 5**

Due Date for “Final Report”. – **May, 3**

“Term Project Presentations” – **May, 4 – May, 6**

As the primary DB-Engine, you are required to use a relational database, such as MySQL, Microsoft SQL Server, or PostgreSQL. These are provided as examples, and your project must be built on a relational database system as the core component. While it is not mandatory, you may explore hybrid database architectures by integrating other database types (e.g., NoSQL, key-value stores) for specific functionalities, such as user authentication or caching. This could provide valuable hands-on experience in designing systems that combine relational and non-relational database technologies.

Project Overview

This term project aims to provide hands-on experience in designing and implementing a database-driven application using a full-fledged **relational database management system (RDBMS)**. Throughout the semester, you will design and implement a database application, which will include an application program (web application or desktop application) and a supporting relational database. The term project includes the following activities spread over the entire semester:

1. Database Design and Development:

- Determine the functionalities and operations of your database application. Think about the various requirements for the user of your application and the various data attributes that need to be stored and later queried (consider factors such as the need to store and query large data volumes, support multiple users, provide concurrent access, maintain consistency, etc.). So, the starting point of your project is to explore questions such as:
 - What is the purpose of the database? What should it do?
 - Who are the users and what are their information needs?
 - What are the problems that the system should solve?
 - What input data is available to the database?
 - What kind of information should be stored in the database?
- Database development:
 - Model the data to be stored in your database, i.e., identify the various entities, relationships, constraints, etc.
 - Design your relational database schema.
 - Write the SQL commands to create your database and tables.

2. Application Development and Demonstration:

- Find appropriate data, and populate the database with at least 5,000 records.
- Develop your application program, which should have supporting codes to access the data from your DBMS.
- Test your system and check if the application works as desired. Your project should "evolve" during the course of the semester, but for the most part, the functionality promised in the first deliverable must be achieved. For this purpose, make sure you perform a thorough requirement analysis at the beginning and balance features so you can "fulfill this contract" in the end.

The end result should be a functioning application that uses your underlying relational database to enable useful functionality.

Project Phases

Phase 1. Database Design and Development

This phase focuses on the conceptual and physical design of the database. It includes the following three steps:

- Phase 1 – Step 1: Requirement Analysis

This first step involves defining the functionalities and operations of the database application. Students should:

- Identify the **main objectives** of their database system.
- Define the **user types** and their respective information needs.
- Specify the **problems the system aims to solve**.
- Determine the **key data attributes and relationships** necessary for supporting queries.
- Ensure that the database **adds value beyond simple data retrieval**, by enabling advanced queries such as: *Monthly sales analysis, customer ranking, inventory forecasting, etc.* Hence, list a number of (**at least five**) meaningful database queries that will be implemented later, in order to show that your application has the potential to have such interesting database queries (*In the final phase we expect to see these advanced queries in your application program*).

- Phase 1 - Step 2: Entity-Relationship (E/R) Design

Based on the requirement analysis, students must create an **E/R diagram** that models:

- Entities, attributes, and relationships.
- Cardinalities and constraints.
- A clear conceptual representation of the database structure.

The E/R diagram should be developed using a suitable tool such as **Visio, Lucidchart, Draw.io, Word**, etc..

- Phase 1 - Step 3: Relational Schema and Database Implementation

The mapping process presented in class is need to be applied to your E/R diagram in order to obtain the relational database schema. Students should:

- Convert the **E/R diagram** into a **relational database schema**.
- Define **SQL CREATE TABLE commands**, including integrity constraints.

Phase 2. Application Development and Final Demonstration

In this phase, students will develop a functional application on top of their designed database. Hence,

- Implement a **fully working software system** that interacts with your database.
- The application should feature:
 - **Graphical User Interface (GUI)** for users.
 - **Login system** with different user types.
 - **CRUD (Create, Read, Update, Delete) operations.**
 - **Advanced queries for reporting & analytics**, showcasing the value of stored data (e.g., complex joins, aggregations, nested queries, etc.).
- **Prepare a Final Report:** Students must submit a Final Report that documents:
 - The **entire project process** (from design to development).
 - The **final database schema** and SQL queries.
 - Screenshots and explanations of **the application interface and functionalities.**
 - **Note:** Submission of the **Final Report is mandatory**. Students who fail to submit **will not be allowed** to present their project.
- **Demonstrate your Work:** Each group will **demonstrate their working system** during the **Term Project Final Demonstration Week.**
 - Each group will have **10 minutes** to showcase their **fully functional system.**
 - **No PowerPoint presentations** are required. Instead, groups must **demonstrate their working software live..**

Term Project - Submission Guidelines

Your work for each phase should be submitted **through the ESTUOYS system** by the specified deadlines.

- **Only one** member of each group should submit the assignment on behalf of the group.
- **Late submissions will not be accepted** after the deadline.

Phase 1 - Submission

1. **Progress Report:** A well-structured report (“word” file) including:

- A cover page (Group No, Project No-Name, Group members, Contribution of the group member)
- Requirement analysis
- E/R diagram (include a picture of your E/R diagram and explain)
- Relational Database Schema (include a picture of your Relational Database Schema inside your report and explain)
- The SQL commands to create your database and tables with some description (You can attach relevant files with .sql extension including commands for data insertion).
- At least 5 interesting/advanced queries

2. **Additional Files**

 **E/R Diagram File** (Editable format: .drawio, .vsdx, .lucid, etc.)

 **SQL Database File (.sql)** – The **database schema**

Phase 2 - Submission

1. **Final Report:** A comprehensive final report (“word” file) describing your work through the semester, including:

- A cover page (Group No, Project No-Name, Group members, Contribution of the group member)
- **Requirement analysis** (*if modified based on Phase 1 feedback – what improvements were made?*)
- **Corrected E/R Diagram** (*if modified based on Phase 1 feedback – what improvements were made?*)
- **Final Relational Database Schema** (*if updated – what improvements were made?*)
- **System Functionalities:** Describe the functionalities of your program with screenshots of your user interfaces (ex: adding new data, deleting some data, modifying some data,

different user types & their access levels, execution results of advanced queries (from Phase-1).

- **Challenges & Improvements:** Any encountered challenges and how they were handled.
- **Conclusion**

2. Project Files

 **Updated E/R Diagram File** (Editable format: **.drawio**, **.vsdx**, **.lucid**, etc.)

 **Application Source Code** (Zipped folder containing **all project files** and necessary dependencies)

 **Database Dump File (.sql)** – Final **database with data** for testing

 **"How to Run" Instructions (how_to_run.txt)** – A step-by-step guide on how to set up and run your application

Final Demonstration Guidelines

- **Mandatory:** Only groups that **submit the Final Report** can participate in the demonstration. Hence, without submitting the Final Report, groups will not be eligible to present their project.
- **Note:** You are only required to demonstrate your program in detail. *Please note that if you do not have a functioning program or code (i.e., if no application has been developed), there will be no content available for demonstration.*